



Laboratory 3

Gradient Descent for Linear Regression

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Overview of exercises

- Laboratory 1: K-Means Clustering (1.0 Point)
- Laboratory 2: Grid Search for Linear Regression (1.0 Point)
- **Laboratory 3: Gradient Descent for Linear Regression (1.5 Point)**
- Laboratory 4: Logistic Regression (1.0 Point)
- Laboratory 5: Basics of Neural Network (1.0 Point)
- Laboratory 6: Image Classification with Neural Network (1.5 Points)
- Laboratory 7: Reinforcement learning (1.5 Point)
- Laboratory 8: Deep RL (1.5 Point)

Dataset

- Consider the following dataset with two input features x_1 and x_2 and target output y .

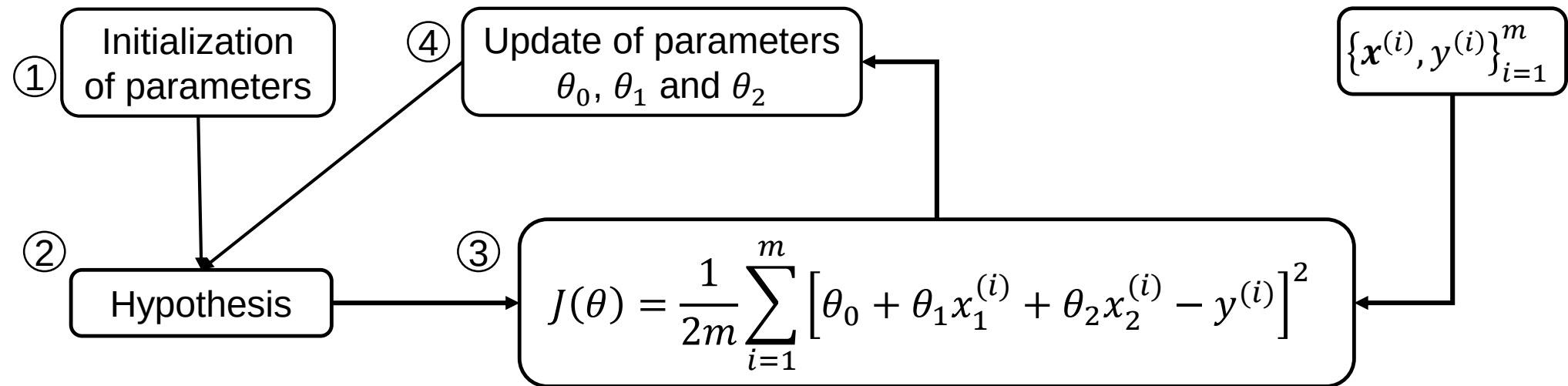
i	x_1	x_2	y
1	1	1	2
2	1	2	3
3	2	1	3

Gradient descent for linear regression

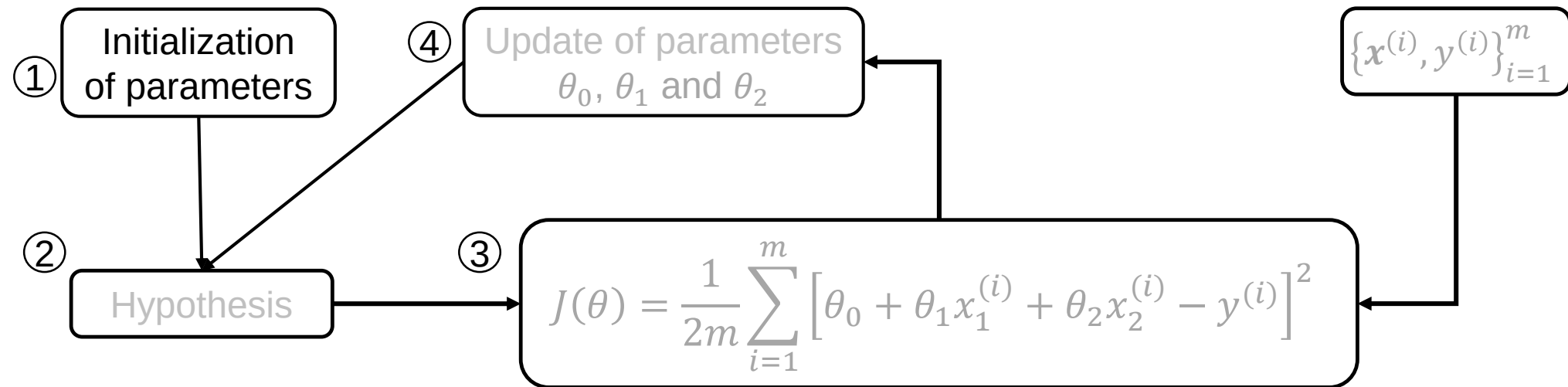
Objective:

- In this example, we fit a linear regression model to the given dataset and calculate its cost function, $J(\theta)$
- We use both batch gradient descent (BGD) and stochastic gradient descent (SGD) for learning the parameters
- To illustrate how the algorithms work, we perform only a single training iteration

Linear Regression Training- Overall training steps



Linear Regression Training- Initialization

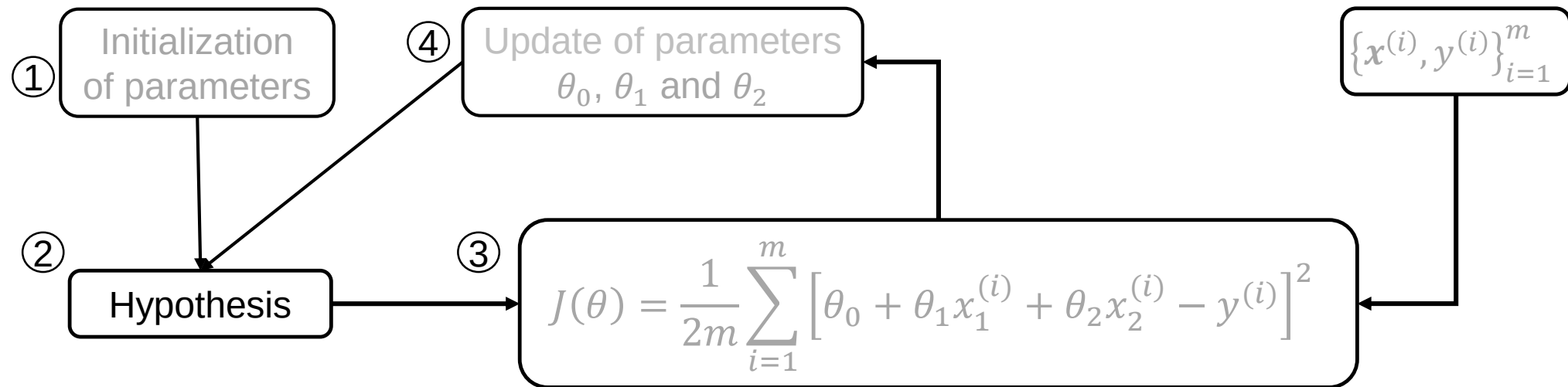


1 Initialization

- We start by initializing the model parameters randomly

$$\theta_0 = \theta_1 = \theta_2 = 1$$

Linear Regression Training- Hypothesis



2 Hypothesis function

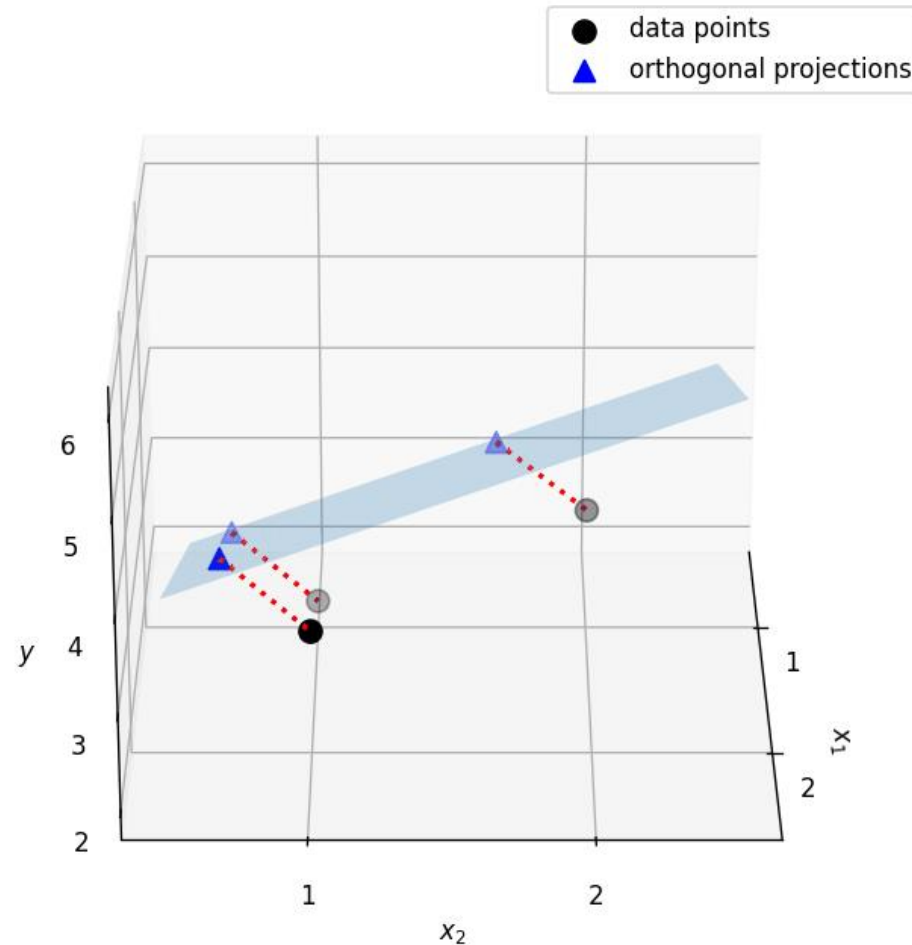
- Consider the hypothesis function as $\hat{y} = h_{\theta}(\mathbf{x}) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$.
- Then the hypothesis function with initial parameter values is

$$\hat{y} = h_{\theta}(\mathbf{x}) = 1 + 1 \cdot x_1^{(i)} + 1 \cdot x_2^{(i)}$$

Visualization of dataset and hypothesis surface

- Errors are shown with red dashed lines

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2, \quad \theta_0 = 1, \theta_1 = 1, \theta_2 = 1$$



Predicted value for first training example

- Compute the predicted values $\hat{y} = h_{\theta}(x^{(i)})$ using the given initial parameters (for all three training examples):

- For first example $i = 1$:

$$\begin{aligned}\hat{y}^{(1)} &= h_{\theta}(\mathbf{x}^{(1)}) = \theta_0 + \theta_1 x_1^{(1)} + \theta_2 x_2^{(1)} \\ &= 1 + 1 \cdot x_1^{(1)} + 1 \cdot x_2^{(1)} \\ &= 1 + 1 \cdot \boxed{1} + 1 \cdot \boxed{1} \\ &= 3\end{aligned}$$

i	x_1	x_2	y
1	1	1	2
2	1	2	3
3	2	1	3

Predicted value for second training example

- For second training example ($i = 2$):

$$\begin{aligned}\hat{y}^{(2)} &= h_{\theta}(\mathbf{x}^{(2)}) = 1 + 1 \cdot x_1^{(2)} + 1 \cdot x_2^{(2)} \\ &= 1 + 1 \cdot 1 + 1 \cdot 2 \\ &= 4\end{aligned}$$

i	x_1	x_2	y
1	1	1	2
2	1	2	3
3	2	1	3

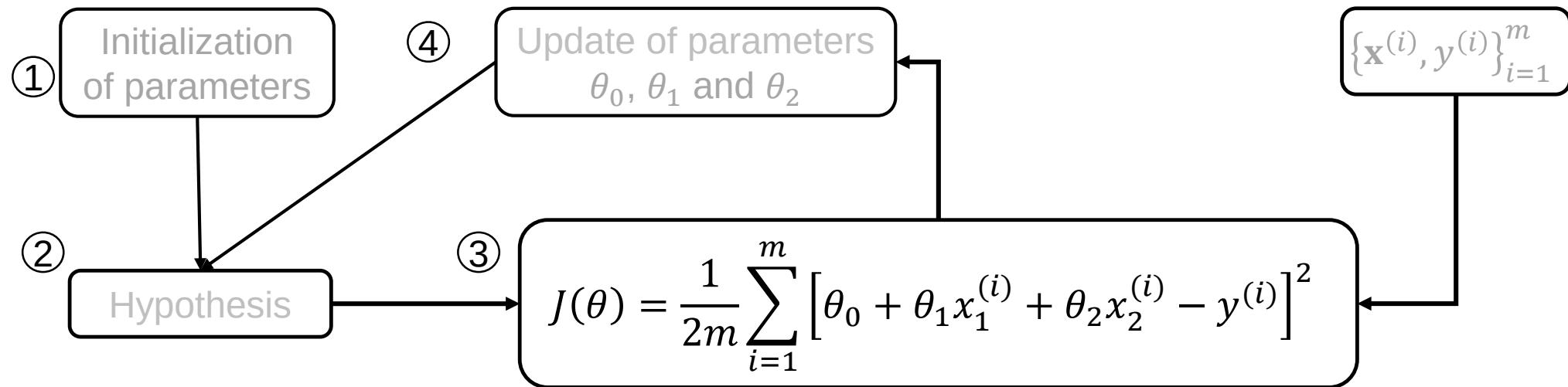
Predicted value for third training example

- For the third training example ($i = 3$):

$$\begin{aligned}\hat{y}^{(3)} &= h_{\theta}(\mathbf{x}^{(3)}) = 1 + 1 \cdot x_1^{(3)} + 1 \cdot x_2^{(3)} \\ &= 1 + 1 \cdot \boxed{2} + 1 \cdot \boxed{1} \\ &= 4\end{aligned}$$

i	x_1	x_2	y
1	1	1	2
2	1	2	3
3	2	1	3

Linear Regression Training- Cost function

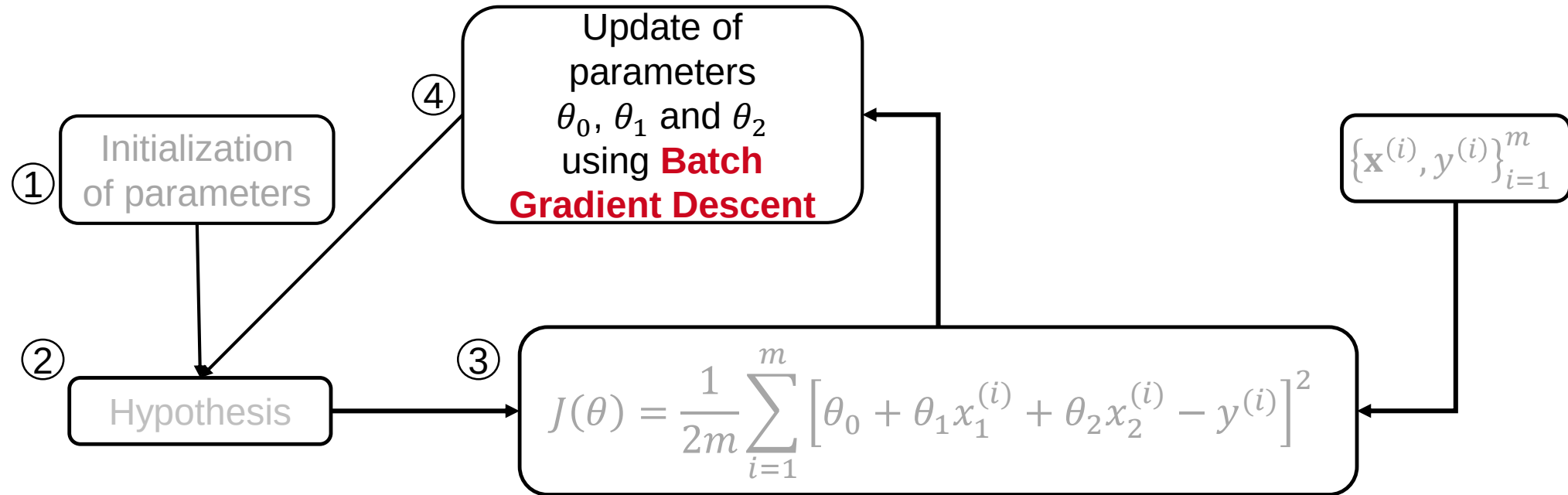


$$\begin{aligned} J(\boldsymbol{\theta}) &= \frac{1}{2m} \sum_{i=1}^{m=3} \left(h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}) - y^{(i)} \right)^2 \\ &= \frac{1}{2m} \sum_{i=1}^{m=3} \left(1 + 1 \cdot x_1^{(i)} + 1 \cdot x_2^{(i)} - y^{(i)} \right)^2 = \\ &= \frac{1}{6} [(3 - 2)^2 + (4 - 3)^2 + (4 - 3)^2] \\ &= 0.5 \end{aligned}$$



Batch Gradient Decent (BGD)

4 Update parameters



- We assume learning rate $\alpha = 0.1$

4 Batch gradient descent, Update parameters

- Batch gradient descent (BGD) uses **all 3 examples** to compute the average gradient.
- Update parameters using BGD:

$$\theta_j \leftarrow \theta_j - \alpha \frac{1}{m} \sum_{i=1}^m [h_{\theta}(x^{(i)}) - y^{(i)}] \frac{\partial}{\partial \theta_j} [h_{\theta}(x^{(i)}) - y^{(i)}]$$

$$\begin{aligned}\theta_0 &\leftarrow \theta_0 - 0.1 \frac{1}{3} \sum_{i=1}^3 [h_{\theta}(x^{(i)}) - y^{(i)}] \\ &= 1 - \frac{0.1}{3} [(3 - 2) + (4 - 3) + (4 - 3)] \\ &= 0.9\end{aligned}$$

i	x_1	x_2	y
1	1	1	2
2	1	2	3
3	2	1	3

$$\begin{aligned}
 \theta_1 &\leftarrow \theta_1 - 0.1 \frac{1}{3} \sum_{i=1}^3 [h_{\theta}(x^{(i)}) - y^{(i)}] x_1^{(i)} \\
 &= 1 - \frac{0.1}{3} [1 \cdot (3 - 2) + 1 \cdot (4 - 3) + 2 \cdot (4 - 3)] \\
 &= 0.86
 \end{aligned}$$

$$\begin{aligned}
 \theta_2 &\leftarrow \theta_2 - 0.1 \frac{1}{3} \sum_{i=1}^3 [h_{\theta}(x^{(i)}) - y^{(i)}] x_2^{(i)} \\
 &= 1 - \frac{0.1}{3} [1 \cdot (3 - 2) + 2 \cdot (4 - 3) + 1 \cdot (4 - 3)] \\
 &= 0.86
 \end{aligned}$$

i	x_1	x_2	y
1	1	1	2
2	1	2	3
3	2	1	3

Hypothesis function with updated parameters

- The hypothesis function with updated parameters is $h_{\theta}(\mathbf{x}) = 0.9 + 0.86x_1 + 0.86x_2$.
- Then the predicted values are

$$\hat{y}^{(1)} = h_{\theta}(\mathbf{x}^{(1)}) = 0.9 + 0.86 \times 1 + 0.86 \times 1 = 2.62$$

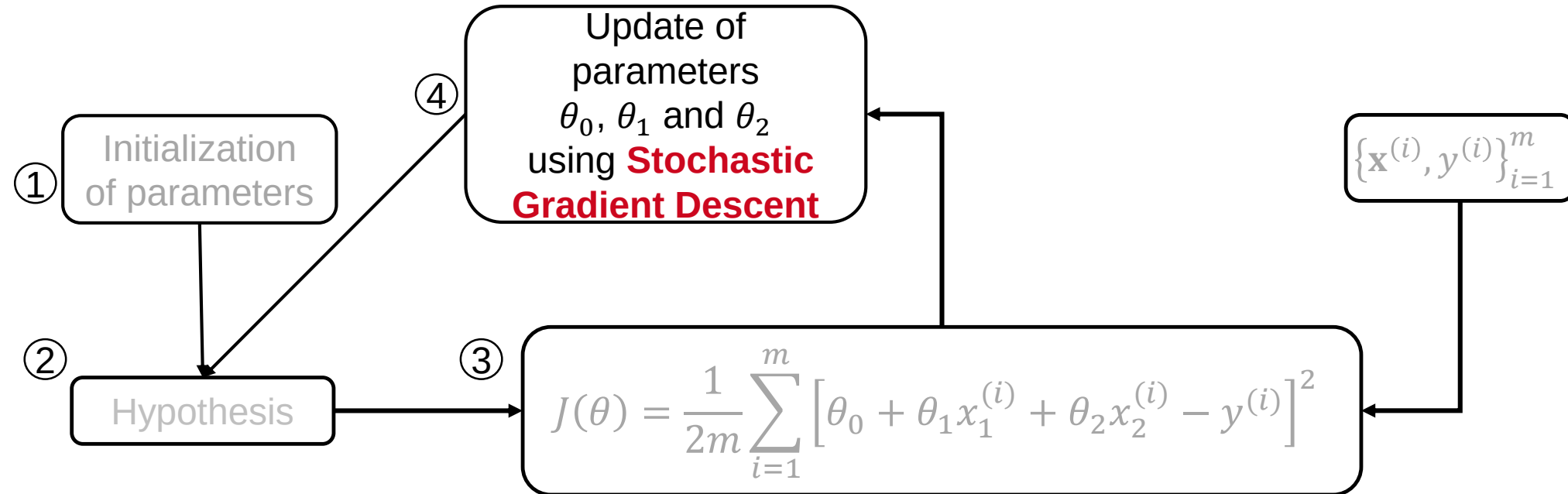
$$\hat{y}^{(2)} = h_{\theta}(\mathbf{x}^{(2)}) = 0.9 + 0.86 \times 1 + 0.86 \times 2 = 3.48$$

$$\hat{y}^{(3)} = h_{\theta}(\mathbf{x}^{(3)}) = 0.9 + 0.86 \times 2 + 0.86 \times 1 = 3.48$$



Stochastic Gradient Decent (SGD)

Linear Regression Training- Update parameters



4 Update parameters

- Stochastic gradient descent update rule is:

$$\theta_j \leftarrow \theta_j - \alpha (h_{\theta}(\mathbf{x}^{(i)}) - y^{(i)}) \frac{\partial}{\partial \theta_j} [h_{\theta}(\mathbf{x}^{(i)}) - y^{(i)}]$$

- Update parameters using the **first** training example ($i = 1$):

$$\begin{aligned}\theta_0 &\leftarrow \theta_0 - 0.1(h_{\theta}(x^{(1)}) - y^{(1)}) \\ &= 1 - 0.1(3 - 2) \\ &= 0.9\end{aligned}$$

i	x_1	x_2	y
1	1	1	2
2	1	2	3
3	2	1	3

$$\theta_1 \leftarrow \theta_1 - 0.1(h_{\theta}(x^{(1)}) - y^{(1)})x_1^{(1)}$$

$$= 1 - 0.1(3 - 2) \times 1$$

$$= 0.9$$

$$\theta_2 \leftarrow \theta_2 - 0.1(h_{\theta}(x^{(1)}) - y^{(1)})x_2^{(1)}$$

$$= 1 - 0.1(3 - 2) \times 1$$

$$= 0.9$$

i	x_1	x_2	y
1	1	1	2
2	1	2	3
3	2	1	3

4 Update parameters, contd.

- Update the parameters using the **second** training example ($i = 2$):

$$\begin{aligned}\theta_0 &\leftarrow \theta_0 - 0.1(h_{\theta}(x^{(2)}) - y^{(2)}) \\ &= \theta_0 - 0.1(0.9 + 0.9x_1^{(2)} + 0.9x_2^{(2)} - y^{(2)}) \\ &= 0.9 - 0.1(3.6 - 3) \\ &= 0.84\end{aligned}$$

i	x_1	x_2	y
1	1	1	2
2	1	2	3
3	2	1	3

$$\theta_1 \leftarrow \theta_1 - 0.1(h_{\theta}(x^{(2)}) - y^{(2)})x_1^{(2)}$$

$$= 0.9 - 0.1(3.6 - 3) \cdot 1$$

$$= 0.84$$

$$\theta_2 \leftarrow \theta_2 - 0.1(h_{\theta}(x^{(2)}) - y^{(2)})x_2^{(2)}$$

$$= 0.9 - 0.1(3.6 - 3) \cdot 2$$

$$= 0.78$$

i	x_1	x_2	y
1	1	1	2
2	1	2	3
3	2	1	3

4 Update parameters, contd.

- Update parameters using the **third** training example ($i = 3$):

$$\begin{aligned}\theta_0 &\leftarrow \theta_0 - 0.1(h_{\theta}(x^{(3)}) - y^{(3)}) \\ &= 0.84 - 0.1(0.84 + 0.84x_1^{(3)} + 0.78x_2^{(3)} - y^{(3)}) \\ &= 0.84 - 0.1(3.3 - 3) \\ &= 0.81\end{aligned}$$

i	x_1	x_2	y
1	1	1	2
2	1	2	3
3	2	1	3

$$\begin{aligned}\theta_1 &\leftarrow \theta_1 - 0.1(h_{\theta}(x^{(3)}) - y^{(3)})x_1^{(3)} \\ &= 0.84 - 0.1(3.3 - 3) \cdot 2 \\ &= 0.78\end{aligned}$$

$$\begin{aligned}\theta_2 &\leftarrow \theta_2 - 0.1(h_{\theta}(x^{(3)}) - y^{(3)})x_2^{(3)} \\ &= 0.78 - 0.1(3.3 - 3) \cdot 1 \\ &= 0.75\end{aligned}$$

i	x_1	x_2	y
1	1	1	2
2	1	2	3
3	2	1	3

Hypothesis function with updated parameters

- The hypothesis function with updated parameters is $h_{\theta}(\mathbf{x}) = 0.81 + 0.78x_1 + 0.75x_2$.
- Then the estimated values are

$$\hat{y}^{(1)} = h_{\theta}(\mathbf{x}^{(1)}) = 0.81 + 0.78 \times 1 + 0.75 \times 1 = 2.34$$

$$\hat{y}^{(2)} = h_{\theta}(\mathbf{x}^{(2)}) = 0.81 + 0.78 \times 1 + 0.75 \times 2 = 3.09$$

$$\hat{y}^{(3)} = h_{\theta}(\mathbf{x}^{(3)}) = 0.81 + 0.78 \times 2 + 0.75 \times 1 = 3.12$$

Assignment

Gradient descent for multivariate linear regression

Objective:

- Implement a multivariate linear regression model to predict the house prices (output labels), given the input features.
- Train the regression model using batch gradient descent (BGD) and stochastic gradient descent (SGD).
- Compare the results and evaluate the model performance on unseen test data.

Tasks

- **Task1.** Implement the multivariate linear regression algorithm, as follows:
Step1. Initialize the parameters. (This function is provided to you.)
Step2. Calculate the predicted values $\hat{y}$ based on hypothesis function
Step3. compute the cost function $J(\theta)$ between the true target values y and the predicted values $\hat{y}$

Note: The input data $\mathbf{x} = [1, x_1, x_2, x_3, x_4, x_5, x_6]$.

Tasks, cntd.

Step4. Implement BGD. Do until loop counter reaches to the max value.

In each iteration:

- Compute predictions for all training examples,
- Compute the (batch) gradient using all training examples,
- Update the parameter vector θ ,
- Record the cost value after at each iteration,
- Return the updated parameters and the list of recorded cost values.

Tasks, cntd.

Step4. Implement SGD. Do until loop counter reaches to the max value.

In each iteration:

- Shuffle the training examples,
- Compute prediction values and errors
- Update the parameter vector θ using one sample at a time,
- Record the cost value after each iteration,
- Return the updated parameters and the list of recorded cost values.

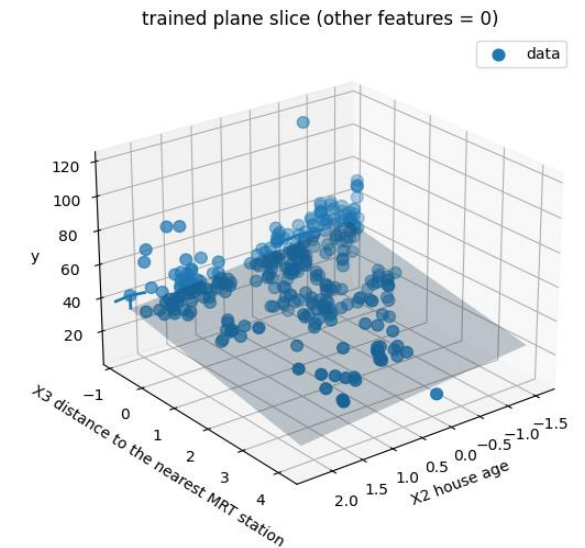
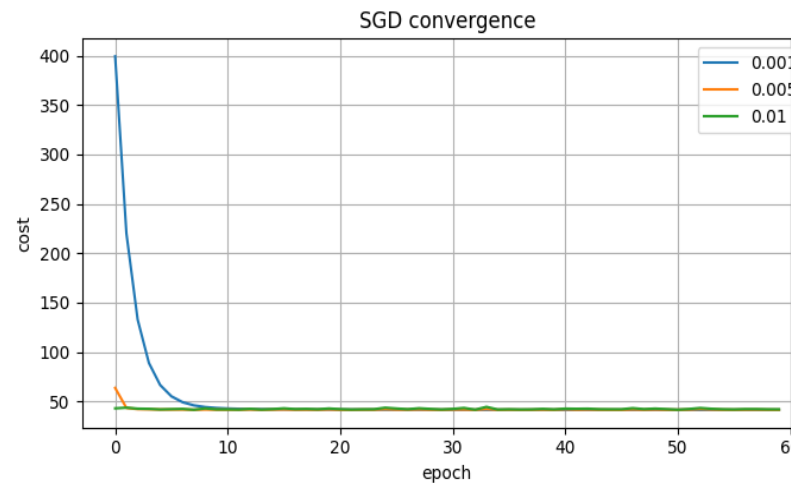
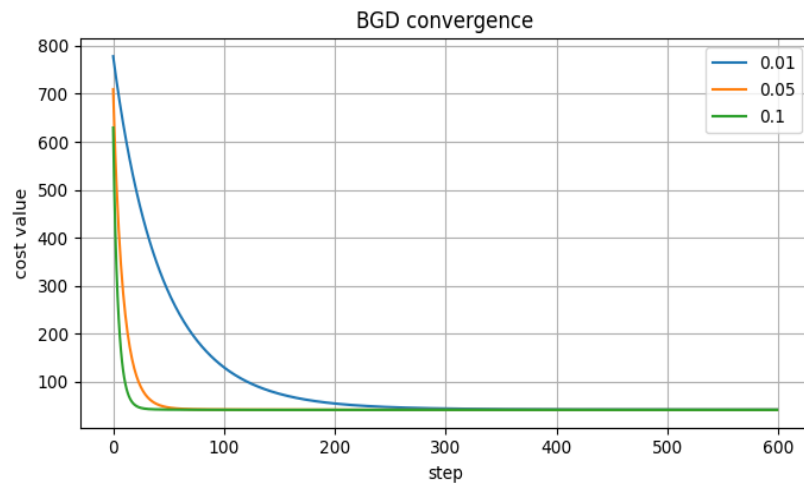
Submission

- Functions you need to implement are (Rest of the code is given):
 - *predicted_values(x,t)*
 - *cost(x, y, t)*
 - *fit_bgd(x, y, lr, steps, seed=0)*
 - *fit_sgd(x, y, lr, epochs, seed=0)*
- Submit your Python code as .py file.
 - If you experience any issues uploading the .py files directly, compress it into a .zip or .tar file and upload the compressed file instead
- Submit your code on Moodle by **May 18, 2026, at 10:30 AM.**
- Do not import any additional libraries or modules that are not already included in the provided code.
- ❖ If the code fails to run, it will receive no points.

Expected result

- Your code should run and plot and print the following results.

Note: Your result may differ a bit from these examples



BGD Cost value for test data: 26.917
SGD Cost value for test data : 27.221

End of Exercise 3